Year 6 Mathematics

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



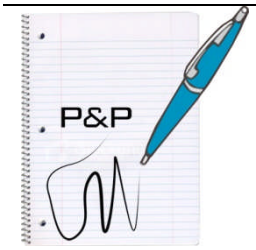
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



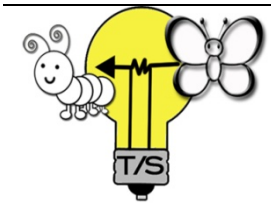
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +\,4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +\,9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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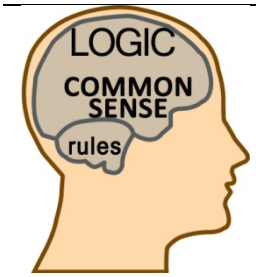
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

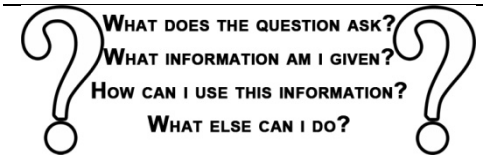
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 6 Mathematics

Number

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Useful formulae and hints

Negative numbers are those that are less than zero. The same operations can be performed on them as positive numbers. As you go along a number line to the left, numbers become less. So

- $-5 < -4$
- The number that is 3 less than -4 is -7,
- The number that is 3 less than 2 is -1, etc

Place value: In the number "abcdefg"

g represents units

f represents tens

e represents hundreds

d represents thousands

c represents tens of thousands

b represents hundreds of thousands

a represents millions

A factor is a number that divides into a given number equally. For example, the factors of 12 are 1, 2, 3, 4, 6 and 12

To answer questions of the type: "8 bricks cost $20, how much are 5 bricks?", reduce to a unit (single) cost. In this case 1 brick costs

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$20 \div 8 = \$2.50$, then multiply by the number you wish to calculate for. In this case, 5 bricks cost $5 \times \$2.50 = \$12.50$

Note the answer can be achieved by the operation $\frac{5}{8} \times \$20$, but ensure you are familiar with the longer method before tackling the short cut

To add fractions with different denominators, change one or both fractions to a common denominator.

If unsure what common denominator to use you can:

- Find the lowest common multiple of the denominators
- Multiply the denominators together (the answer may then require simplification)

Example $\frac{1}{3} + \frac{5}{6} = \frac{2}{6} + \frac{5}{6} = \frac{7}{6}$

Example $\frac{1}{3} + \frac{3}{4} = \frac{4}{12} + \frac{9}{12} = \frac{13}{12}$

Example $\frac{3}{4} + \frac{1}{6} = \frac{18}{24} + \frac{4}{24} = \frac{22}{24} = \frac{11}{12}$

To change a fraction to a percentage, multiply the fraction by 100

To change a percentage to a fraction, divide the number by 100

To change a fraction to a decimal, divide the numerator by the denominator

To change a decimal to a fraction, divide the number (without the decimal point) by 10, 100, 1000, etc where the number of zeroes is equal to the number of figures after the decimal point. Simplify the fraction if necessary

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Examples: $0.2 = \frac{2}{10} = \frac{1}{5}$

$0.11 = \frac{11}{100}$

$0.255 = \frac{255}{1000} = \frac{51}{200}$

To find a percentage of a number, change the percentage to a fraction as above, and multiply the number by this fraction

Example: $15\%\ of\ 200 = \frac{15}{100} \times 200 = \frac{3000}{100} = 3000$

Order of operation is brackets, indices, multiplication, division, addition, subtraction

A prime number is one that has only two factors; itself and 1 (note 1 is NOT a prime number)

A composite number is one that has more than two factors

To add or subtract decimals, line up the two numbers according to their decimal points, then add or subtract as normal, carrying the decimal point down to the same place in the answer

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Exercise 1

Negative Numbers

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1) Complete the following patterns

a) 5, 4, 3, ____, ____, _____

b) 6, 4, 2, ____, ____

c) 9, 6, 3, ____, ____

d) 10, 5, ____, ____

e) 6, 2, ____, ____

f) 5, 3, 1, ____, ____

g) 13, 7, 1, ____, ____

2) Place the following numbers on a number line

a) 10

b) 6

c) --1

d) 1

e) 0

f) –2

g) –5

h) –10

i) –1.5

3) Calculate the following

a) $6 - 6$

b) $4 - 4$

c) $4 - 5$

d) $3 - 7$

e) $4 - 10$

f) $0 - 3$

g) $0 - 5$

4) Calculate the following

a) $-2 - 3$

b) $-1 - 4$

c) $-5 - 2$

d) $-10 - 5$

e) $-3 - 12$

5) Calculate the following

a) $-4 + 4$

b) $4 - 4$

c) $-2 + 4$

d) $4 - 2$

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e) $-1+3$

f) $3-1$

g) $-10+9$

h) $9-10$

i) $-8+3$

j) $3-8$

6) From your answers to question 5, rewrite the following question and hence make it easier to solve

$$-a+b$$

Where a and b are any numbers

7) The temperature at two o'clock in the afternoon was 10 degrees. Over the next 3 hours the temperature dropped by 12 degrees. What was the temperature at 5 o'clock?

8) Canasta is a card game where at the end of each hand points are either added or deducted from a player's score. Players' scores start at zero. In the first hand Tom lost 25 points, and in the second he lost 15. What was his score after 2 hands?

9) Ian had $40 in his bank account. He wrote a cheque for $50 for petrol. Assuming his bank let the payment go through, what was the balance of Ian's account after he paid for the fuel?

10) In golf, every hole has a "par score" which is the amount of shots it usually takes to get the ball into the hole. Each player records how many shots it actually took, and the total score above or below "par" is recorded. The following score card shows Peter's score for 9 holes of golf.

Hole Number	Par Score	Peter's Score	Amount above or below par	Running Total
1	5	4	--1	--1
2	4	3	--1	--2
3	4	5	+1	--1
4	3	2		
5	5	5		
6	4	3		
7	4	3		

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8	3	5		
9	5	4		

Complete the table

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Exercise 2

Place Value

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1) Write the following in numerals

a) Ten thousand three hundred and twenty seven

b) Forty two thousand two hundred and twelve

c) Seven thousand and seven

d) Sixteen thousand and fifteen

e) One hundred and twelve thousand four hundred and twenty

f) One hundred and two thousand and eleven

2) Write the following in words

a) 32033

b) 410002

c) 11706

d) 105007

e) 310207

f) 190001

3) What is the place value of the 5 in each of the following?

a) 21005

b) 151443

c) 526111

d) 1523123

e) 10054

f) 500121

4) Write the following numbers in order, from largest to smallest

2121234, 211246, 213652, 100834, 1000999, 2221011, 2021101,

5) Write the following numbers in order, from smallest to largest

104224, 14425, 21501, 115001, 100516, 101010, 110110, 151002, 105509

6) There were 126244 people at a soccer match. Write this number to the nearest

a) Hundred

b) Thousand

c) Ten thousand

7) Round the number 167532556 to the nearest:

a) Ten

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b) Hundred

c) Thousand

d) Ten thousand

e) Hundred thousand

f) Million

8) Add the following

a) 3270 + five hundred and seventy five

b) Two thousand and nine + 1747

c) One hundred and twenty thousand one hundred + eighteen thousand two hundred and twelve

d) 19143 + three thousand one hundred and two

e) 17111 + ninety nine

9) Which numeral represents hundreds in the number 323068?

10) If 500,000 is added to the number 1686,400, which numerals change place value?

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Exercise 3

Operations on Whole Numbers

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1) Add the following

a) 254 + 236

b) 177 + 427

c) 281 + 945

d) 139 + 553

e) 336 + 624

f) 172 + 228

2) Subtract the following

a) $499 - 544$

b) $813 - 632$

c) $167 - 146$

d) $710 - 511$

e) $884 - 193$

f) $757 - 459$

3) Add the following

a) 293 + 682

b) 164 + 461

c) 373 + 513

d) 1112 + 1030

e) 2146 + 119

f) 1637 + 104

4) Subtract the following

a) $2174 - 162$

b) $3012 - 153$

c) $4292 - 113$

d) $8162 - 333$

e) $7313 - 618$

f) $9001 - 565$

5) Multiply the following

a) 422×5

b) 336×8

c) 7×525

d) 111×13

e) 273×12

f) 531×15

6) Multiply the following

a) 34×272

b) 52×289

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c) 161×22

d) 503×41

e) 665×37

f) 711×19

7) Divide the following

a) $88 \div 9$

b) $82 \div 7$

c) $51 \div 6$

d) $73 \div 12$

e) $61 \div 4$

f) $73 \div 8$

8) Divide the following

a) $150 \div 10$

b) $2200 \div 10$

c) $1800 \div 20$

d) $2310 \div 10$

e) $147 \div 10$

f) $222 \div 10$

9) Divide the following

a) $1200 \div 100$

b) $6200 \div 100$

c) $3210 \div 100$

d) $2591 \div 100$

e) $1111 \div 100$

10) Calculate the following using the correct order of operation

a) $3 \times 6 + 9$

b) $11 \times 7 + 3$

c) $10 - 8 \div 4$

d) $16 \div 4 + 4$

e) $3 \times 6 \div 3$

f) $10 + 100 \div 10$

11) Determine if the following are prime or composite by analysing their factors

a) 4

b) 11

c) 12

d) 100

e) 23

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f) 45

g) 51

h) 77

i) 79

j) 1243

12) A supermarket is having a special on milk. The usual price is $3 per carton, but the special price is 3 for $7.50. How much **per carton** are shoppers saving?

13) The price of steak is $22.50 per kilogram. How much is this per 100 grams?

14) Peter walked 8km. Dan walked twice as far plus 5 km. How far did Dan walk?

15) Which is the better buy and by how much per carton? Five cartons of eggs for $21.05, or 7 cartons of eggs for $30.10

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Exercise 4

Fractions & Percentages

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1) Add the following fractions

a) $\frac{1}{2}+\frac{3}{4}$

b) $\frac{3}{8}+\frac{1}{4}$

c) $\frac{1}{5}+\frac{3}{10}$

d) $\frac{1}{3}+\frac{5}{6}$

e) $\frac{3}{5}+\frac{1}{10}$

f) $\frac{5}{6}+\frac{1}{12}$

2) Subtract the following fractions

a) $\frac{1}{2}-\frac{1}{6}$

b) $\frac{4}{5}-\frac{1}{10}$

c) $\frac{1}{2}-\frac{1}{12}$

d) $\frac{1}{10}-\frac{3}{100}$

e) $\frac{3}{4}-\frac{1}{8}$

f) $\frac{1}{5}-\frac{11}{100}$

3) Calculate the following

a) $3\times\frac{1}{5}$

b) $\frac{2}{3}\times 4$

c) $\frac{5}{12}\times 2$

d) $5\times\frac{3}{8}$

e) $\frac{3}{8}\times 2$

f) $\frac{3}{100}\times 11$

g) $\frac{7}{10}\times 3$

4) Calculate the following

a) $\frac{1}{2}\times 10$

b) $\frac{1}{3}\times 21$

c) $20\times\frac{1}{4}$

d) $35\times\frac{1}{5}$

e) $\frac{1}{6}\times 42$

f) $\frac{1}{8}\times 72$

g) $130\times\frac{1}{10}$

h) $\frac{1}{100}\times 130$

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5) Write the following as percentages

a) $\frac{1}{2}$

b) $\frac{1}{5}$

c) $\frac{1}{4}$

d) $\frac{1}{10}$

e) $\frac{1}{8}$

f) $\frac{1}{100}$

6) Calculate the following

a) $10\% \ of \ 120$

b) $25\% \ of \ 200$

c) $50\% \ of \ 360$

d) $20\% \ of \ 120$

e) $50\% \ of \ 10\% \ of \ 100$

f) $25\% \ of \ 20\% \ of \ 200$

7) John eats one-third of a cake and Peter eats one-sixth. What fraction of the cake has been eaten?

8) On day one, Martin read one quarter of a book. On the second day he read three eighths of the book. On the third day he finished the book. What fraction of the book did he read on the third day?

9) Bill bought one fifth of all the apples left on a shelf. If he bought 8 apples, how many were left on the shelf?

10) John ate 25% of a cake and Bill ate one quarter. Who ate more?

11) At a sale Mary got 50% off the price of a dress that originally cost $40. Sally got 10% off a dress that originally cost $300

a) Who saved most money?

b) Who spent the most and by how much?

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12) In a mathematics test Tom got $\frac{1}{4}$ of the questions wrong, and Alan got $\frac{1}{3}$ of the questions wrong. If there were 60 questions on the test who got more questions correct and by how many?

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Exercise 5

Decimals

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1) Express the following as decimals

a) $\frac{7}{1000}$

b) $\frac{3}{1000}$

c) $\frac{23}{1000}$

d) $\frac{1}{1000}$

e) $\frac{100}{1000}$

2) Express the following as fractions

a) 0.002

b) 0.035

c) 0.014

d) 0.001

e) 0.033

3) Add the following

a) 0.231 + 0.42

b) 0.15 + 0.162

c) 0.313 + 0.45

d) 0.071 + 0.28

e) 0.55 + 0.405

f) 0.83 + 0.03

4) Add the following

a) 0.518 + 0.36

b) 0.75 + 0.108

c) 0.222 + 0.69

d) 0.054 + 0.87

e) 0.919 + 0.51

f) 0.86 + 0.418

5) Add the following

a) 1.421 + 2.11

b) 1.651 + 0.22

c) 2.35 + 1.219

d) 4.203 + 1.62

e) 5.171 + 3.11

f) 1.55 + 1.586

6) Subtract the following

a) 0.514 – 0.23

b) 0.86 – 0.135

c) 0.929 – 0.48

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d) 0.777 – 0.616

e) 0.125 – 0.02

f) 0.215 – 0.24

7) Subtract the following

a) 1.41 – 0.621

b) 1.89 – 0.922

c) 2.12 – 0.463

d) 3.24 – 2.356

e) 9.57 – 7.194

f) 2.15 – 0.995

8) Calculate the following

a) $0.125 \div 5$

b) 0.14×5

c) $1.44 \div 8$

d) $3.5 \div 5$

e) 0.22×6

f) 0.15×8

g) $2.22 \div 3$

h) $1.5 \div 5$

9) Calculate the following

a) $2.34 \div 10$

b) 0.15×100

c) 1.25×1000

d) 0.03×10

e) $225.4 \div 100$

f) $1122.33 \div 1000$

8) Tom had \$15.20, whilst Alan had ten times as much. How much money did Alan have?

9) The price of petrol is five times more than it was ten years ago. Today's price is \$1.40 per litre. How much was petrol per litre 10 years ago?

10) After midnight the owner of a 24 hour store triples his prices. How much do each of the following cost after midnight?

a) A loaf of bread costs \$1.80 before midnight

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b) A bottle of cool drink costs $2.15 before midnight

c) A burger costs $3.10 before midnight

d) A chocolate bar costs 95 cents before midnight

11) A hot dog costs $4.65 after midnight in the store from question 10. How much does it cost before midnight?

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Year 6 Mathematics

Chance & Data

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Useful formulae and hints

The chance of an event happening range from 0 (impossible) to 1 (certain). A chance of ½ represents an event where there are two possible outcomes and each is as likely to occur as the other (Tossing a coin)

The probability of an event happening $= \frac{Number\ of\ desired\ outcomes}{Total\ possible\ outcomes}$

Example: the probability of rolling a number greater than 4 on a six sided die is $\frac{2}{6}$ since there are 6 possible outcomes, 2 of which are desired. Note this fraction can then be simplified.

To calculate the section of a pie graph corresponding to a figure on a table,

- Divide the figure by the total of the table
- Multiply this figure by 360
- This gives the number of degrees in the section

Example:

Colour	Number
Red	6
Blue	4
Green	2
Total	12

For red, $\frac{6}{12} \times 360 = 180°$

For blue, $\frac{4}{12} \times 360 = 120°$

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For green, $\frac{2}{12} \times 360 = 60°$

Ensure that the sum of all degrees is 360

The mean of a set of data $= \frac{Sum\ of\ scores}{Number\ of\ scores}$

Example, for the data set 2, 4, 4, 10, the sum of the data is 20, the mean $= \frac{20}{4} = 5$

Note the mean does not have to be a number in the data set

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Exercise 1

Assigning Probabilities

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1) For each of the following events

- State as a fraction the chance of the named event occurring
- Express this fraction as the simplest equivalent fraction
- Express the chance as a percentage
- (e.g. $\frac{3}{6} = \frac{1}{2} = 50\%$)

a) You spin white on a spinner with five white and five black sections

b) You pull a black shirt out of a draw containing 2 black and 6 white shirts

c) You roll a prime number on a six sided dice

d) A bag contains 25 balls numbered from 1 to 25; you pull out a ball whose number is divisible by 5

e) A bag contains balls numbered from 1 to 100. You pull out a ball that has a number more than 19 and less than 30 on it.

2) Place the probabilities from question 1 on a number line

3) Draw a spinner that has three colours; red, blue and green; and each colour has an equal probability of being spun

4) Draw a spinner that has three colours; red, green and blue; so that blue has twice as much chance of being spun as red or green.

5) Draw the six faces of a dice for which there is an equal chance of rolling the number 1, 2, or 3, but no chance of rolling any other number

6) Draw the six faces of a dice for which there is more chance of rolling a one than 2, 3, 4, or 5

7) There are 200 cubes in a bag, some white and some black. You draw out 100 of them and notice there are 52 black and 48 white. What could you say about the numbers of black and white cubes left in the bag

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8) You have to put ten shirts into a draw, some red some white and some black. How many black shirts do you have to put in the draw to make sure you have a better chance of pulling one out than any of the other colours?

9) Design a dartboard with 4 numbers on it so that you have a one in two chance of hitting the number 10, a one in four chance of hitting the number 20, and a one in eight chance of hitting the numbers 30 or 40

10) Draw a map of an imaginary world. If you close your eyes and point to a spot on your map there will be twice as much chance of hitting land as there is hitting water

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Exercise 2

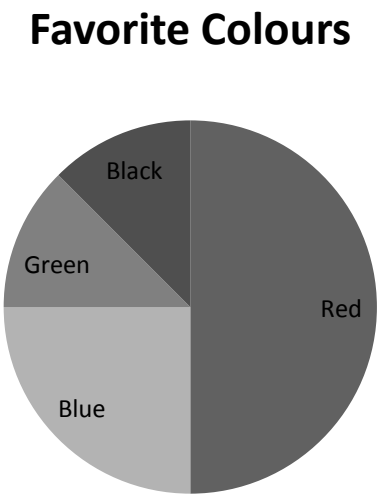
Pie Graphs

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1) Complete the following table

Category	Quantity	Percentage	Degrees of pie chart
1	10	5%	
2	40		
3	20		36
4	50		
5	80		
	200	100	360

2) The pie graph below shows the favourite colour of a number of people



a) Sixty people's favourite colour was red. How many people were surveyed?

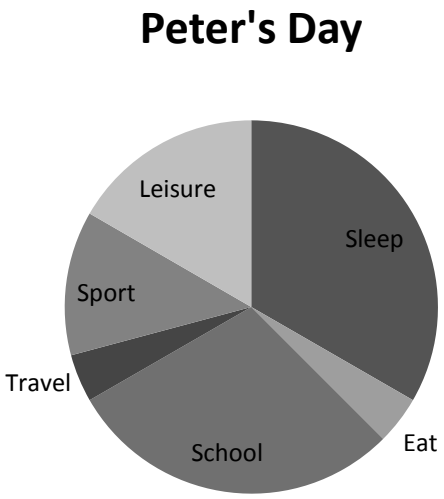
b) How many people preferred blue?

c) Which two colours were preferred by the same number of people?

d) Name a colour that was nobody's favourite?

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3) The pie graph shows the amount of time Peter spends doing various things in a 24 hour day



a) Which activities does he spend least time on?

b) What fraction of the circle is taken up by sleep?

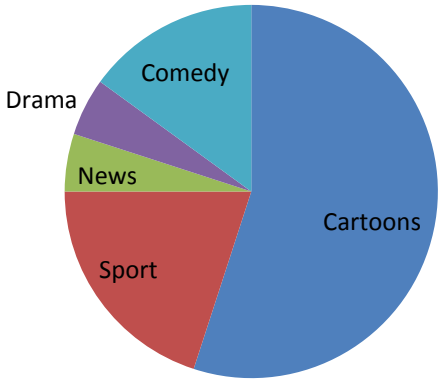
c) How many hours of sleep does Peter get?

d) Does he spend more time on sport or leisure? Explain your answer

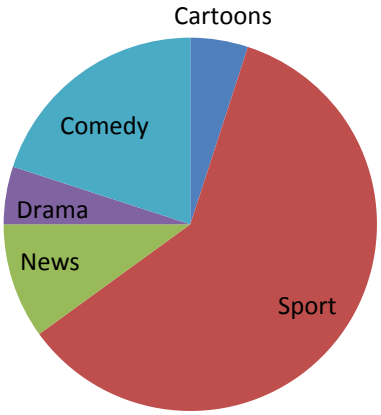
4) Alan constructed a pie chart when he was ten years old which showed how much time he spent watching certain types of TV programs. He then constructed a similar chart when he was twenty years old

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TV Viewing at ten years of age



TV Viewing at twenty years of age



a) Which type of show had the biggest increase in Alan's viewing time?

b) Which type of show had the biggest decrease in Alan's viewing time?

c) For which type of show did Alan's viewing habits change the least?

d) What percentage of Alan's time was spent watching news when he was ten, and what percentage does he spend now?

e) Explain how the pie graph might change if Alan was to draw one when he was sixty years old

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5) Draw a pie chart that reflects the following information

People's favourite sport

Sport	Number of people
AFL	60
Rugby	20
Netball	40
Tennis	30
Golf	30

6) A class took a survey of each student's favourite fruit and drew the following graph from their results. One piece of fruit equals one vote



Display this information in a pie chart

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7) Discuss why a pie chart may not be suitable for the following data

WEEK NUMBER	NUMBER OF RAINY DAYS
1	2
2	4
3	0
4	6
5	7
6	4
7	5
8	3
9	2
10	0

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Exercise 3

Mean

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1) Find the mean of the following

a) 2, 2, 2, 2, 2, 2

b) 1, 2, 3, 4, 5

c) 2, 4, 6, 8, 10

d) 10, 20, 30, 40, 50

e) 100, 200, 300, 400, 500

2) Find the mean of the following

a) 5, 6, 7, 8, 9

b) 5, 6, 7, 8, 9, 1000

c) 100, 102, 105, 107, 110

d) 1, 100, 102, 105, 107, 110

3) From your answers to question 2, what effect does adding a number greater or less than the rest of the data have on the mean?

4) The mean of a set of data is 5. The data is:

1, 3, 7, 8, 9 and an unknown number

What is the unknown number?

5) The mean of the following set of data is 10:

2, 8, 12, 15 and an unknown number

What is the unknown number?

6) Alan played 5 games of basketball and recorded how many points he scored. His scores were:

12, 6, 22, 4, 26

What was his average score per game?

7) In his next 5 games Alan averaged 10 points per game. Does this mean that in at least one of his games Alan scored exactly 10 points? Explain your answer

8) There are 10 boys and 10 girls in the year six maths class. On a test the boys' average score was 70 and the girls' average score was 80. What was the average score for the class?

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9) There are 10 boys and 15 girls in the year seven maths class. On a test the boys' average score was 60 and the girls' average score was 80. What was the average score for the class?

10) What happens to the mean of a set of data when every score is multiplied by two? Explain using examples

11) What happens to the mean of a set of data when five is added to every score? Explain using examples

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Year 6 Mathematics

Algebra & Patterns

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Useful formulae and hints

When solving word problems, follow a systematic approach

- Let the unknown number be represented by a blank space
- Write a number sentence from the word sentence
- Fill in the missing number

Example, twice a certain number plus 3 is equal to eleven. What is the number?

$2 \times \blacksquare + 3 = 11$

$2 \times \blacksquare = 8$

$\blacksquare = 4$

When finding the rule for a pattern, follow this procedure

Look for arithmetic difference between two terms and see if that applies to all terms in the sequence

Example

1, 3, 5, 7...

The difference between the first two terms is 2

The difference between each pair of terms is 2

The pattern has been found; **add two to each term**

If the pattern is not arithmetic, look for a more complicated pattern

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Example

1, 3, 7, 15...

The difference between the first two terms is 2

The difference between the next two terms is 4

The pattern has not been found

If you multiply the first term by 2 then add 1, you get the second term

2 x 3 + 1=7

7 x 2 + 1 = 15

The pattern has been found

Example

1, 5, 17, 53

If you multiply the first term by 2 and add 3, you get the second term

5 x 2 + 3 = 13, the pattern has not been found

If you multiply the first term by 3, and add 2 you get the second term

5 x 3 + 2 = 17

17 x 3 + 2 = 53

The pattern has been found

If the pattern is still not found, repeat for higher numbers

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Exercise 1

Calculating Values

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1) Calculate the missing values in the following

a) $3 + 7 = 12 - \blacksquare$

b) $8 + 9 = 20 - \blacksquare$

c) $5 + 11 = 25 - \blacksquare$

d) $22 - 8 = 11 + \blacksquare$

e) $\blacksquare - 7 = 21 - 9$

f) $6 + \blacksquare = 15 - 3$

2) Calculate the missing values in the following

a) $42 \div 7 = 3 \times \blacksquare$

b) $24 \div 3 = \blacksquare \times 8$

c) $6 \times 5 = \blacksquare \div 3$

d) $4 \times 10 = 80 \div \blacksquare$

e) $1 \times 1 = 8 \div \blacksquare$

f) $12 \times \frac{1}{2} = 12 \div \blacksquare$

3) Calculate the missing values in the following

a) $4 \times 3 = \blacksquare - 6$

b) $7 \times \blacksquare = 14 + 7$

c) $22 \div 11 = 12 - \blacksquare$

d) $24 \div \blacksquare = 1 \times 4$

e) $\blacksquare \times \blacksquare = 4 + 5$

f) $300 \times \blacksquare = 45 - 45$

4) Peter thinks of a number and doubles it. After this it is equal to John's number plus 8. If John's original number was 14, what was Peter's original number?

5) Alan's dad says to Alan "If you take your age now and triple it, it will be equal to my age now plus three". If Alan is 12 years old, how old is his dad?

6) Greg runs a lap of the track in 4 minutes. In the time it takes him to run three laps, Tom runs two laps. How long does it take Tom to run a lap?

7) If you double the number of behinds a football team kicked and subtract three from the number of goals it kicked, you get the same number. If they kicked 21 goals how many behinds did they kick?

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8) Ben's mother cut up a block of chocolate into equal sized pieces. She cut it so that Ben and his seven friends each got the same amount of chocolate, and there were 2 pieces left over. If each of the children got 4 pieces, how many pieces did she cut the chocolate bar into?

9) If you multiply the temperature in degrees Celsius by 9, then divide the result by five, the result is the temperature in Fahrenheit minus 32. If the temperature is 40 degrees Celsius, what is the temperature in Fahrenheit?

10) One third of light globes in a carton were broken. Of the number left there was enough to put two lights in each of five rooms. How many light globes were originally in the carton?

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Exercise 2

Number Patterns

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1) For the following series, fill in the next two terms

a) 1, 3, 7, 15

b) 2, 3, 5, 9

c) 1, 4, 13, 40

d) 3, 10, 24

2) For the following series, fill in the next two terms

a) 62, 30, 14

b) 32, 44, 68, 116

c) 100, 60, 40,

d) 60, 32, 18

3) Fill in the blanks in the following

a) 2, 7, ___, 67, 202, ___

b) ___, 2, 6, 22, 86, ____

c) 1, 2, 5, 14, ___

d) 0.5, 1.5, 3.5, ___, ___

e) $\frac{1}{2}, \frac{1}{5}$, ___, $\frac{1}{23}$, ___

4) What are the next three numbers of the following series?

1, 3, 6, 10

5) Each day Peter decided to walk double the number of steps he had taken the day before plus 10 extra steps. On the first day he walked 10 steps

a) How many steps did he take on day 2?

b) How many steps did he walk on day 5?

c) How many steps had he taken at the end of the week?

6) On week one of his diet, George lost .2 kg. Each week after that he lost double the previous weeks minus 1kg.

a) How much weight did he lose in week 3?

b) After five weeks he had reached his goal weight. What was his average weight loss per week?

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7) A pond can only hold 100 fish until it has to be cleaned out. In the first month there were 4 fish in the pond. Each month the number of fish in the pond was equal to three times the previous months' number less 6. During which month was it necessary to clean out the fish pond, and how many fish would have been in the pond at the end of that month if it had not been cleaned out?

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Year 6 Mathematics

Measurement:

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Useful formulae and hints

Distance = Speed x time

100 km per hour for 3 hours = 100 x 3 = 300 km

Average speed = Distance ÷ time

400 km in 8 hours = 50 km per hour average speed

Time = Distance ÷ speed

500 km at an average speed of 50 km per hour = 10 hours

The perimeter of a shape is the distance around its outside

The area of a rectangle or square is equal to length x width

The area of a triangle is equal to the length of the base x the perpendicular height, then halved

The area of a trapezium is (half the sum of the two sides), multiplied by the height

1 cm^3 displaces 1 millilitre of liquid

1000 cm^3 displace 1 litre of liquid

There are 1000 grams in 1 kg

To convert from g to kg, divide by 1000

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4200 g = 4.2 kg

To convert from kg to g, multiply by 1000

3.75 kg = 3750 g

To calculate difference in time when moving from one hour to another:

Find the number of minutes to complete the first hour

Add the number of minutes in the second hour

Example: find the time between 4:36 and 5:12

From 4:36 to 5 o clock is 24 minutes

24 + 12 = 36 minutes

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Exercise 1

Distance & Travel

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1) Calculate the distance around the outside of a square that has a side length of

a) 3 mm

b) 8 cm

c) 32 m

d) 5.4 km

e) 70 cm

2) Calculate the distance around the outside of a rectangle that has side lengths of

a) 4m and 3 m

b) 17mm and 10mm

c) 4 m and 50cm

d) 0.8 km and 60cm

e) 90 mm and 9cm

3) Calculate the distance around the outside of an equilateral triangle of side length

a) 9 cm

b) 3 m

c) 1.2c m

d) 4 km

e) 1.2 m

4) Calculate the distance around the outside of an isosceles triangle with the following measurements

a) 10cm for the common sides, and 5cm for the unique side

b) 100 cm for the common sides, and 5cm for the unique side

c) 10cm for the common sides, and 15cm for the unique side

d) 30cm for the common sides, and 8cm for the unique side

e) 25mm for the common sides, and 1cm for the unique side

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5) A car travels at an average speed of 100 km per hour. How long would it take to complete a journey of 300 km?

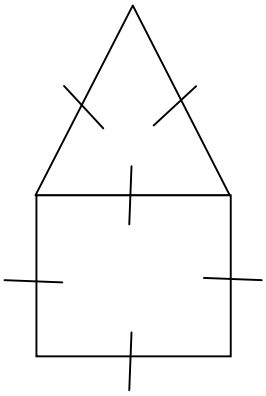
6) A man walks at an average speed of 5 km per hour. How long would it take him to walk 7.5 km?

7) How long would it take an ant that walks at an average speed of 2cm per minute to walk around a square of side length 5cm?

8) An equilateral triangle has a side length of 8 cm. It took a snail one hour to walk around the triangle. What was its average speed?

9) A rectangle has one side that measures 45 mm and another that measures 2cm. How long would it take to go around its perimeter if the average speed was 1.3 cm per minute?

10) How long would it take a man walking at 2 km per hour to go around the outside of this shape?



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Exercise 2

Area

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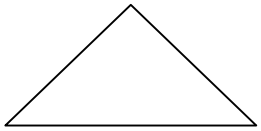
Use the following scale to calculate the area of the following shapes

---3 cm---

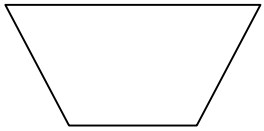
1)



2)

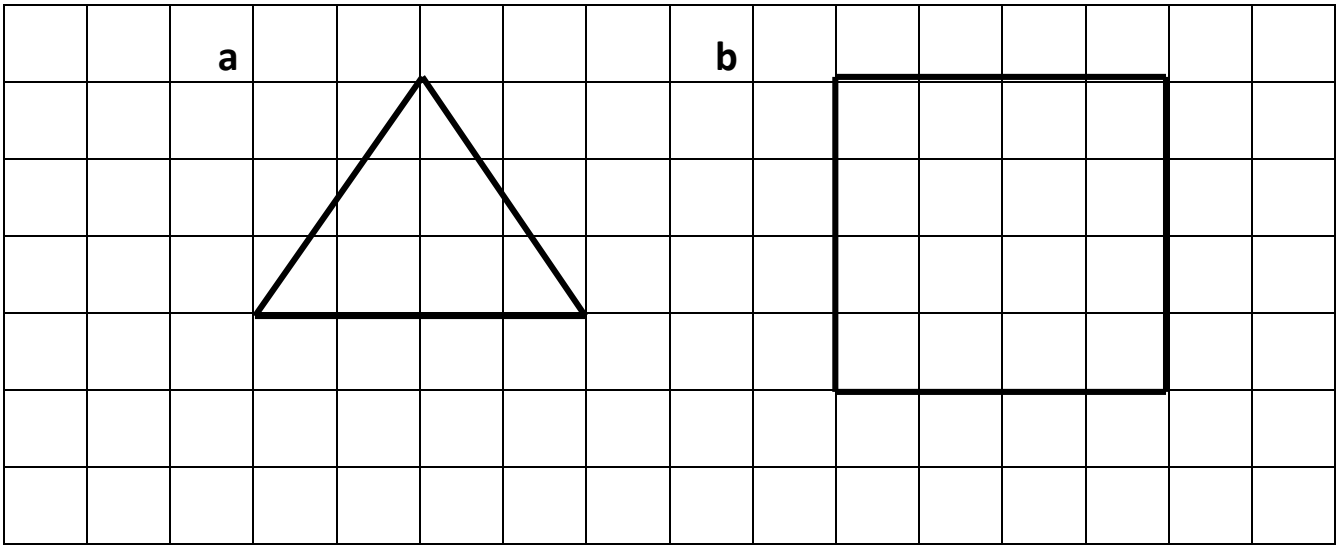


3)

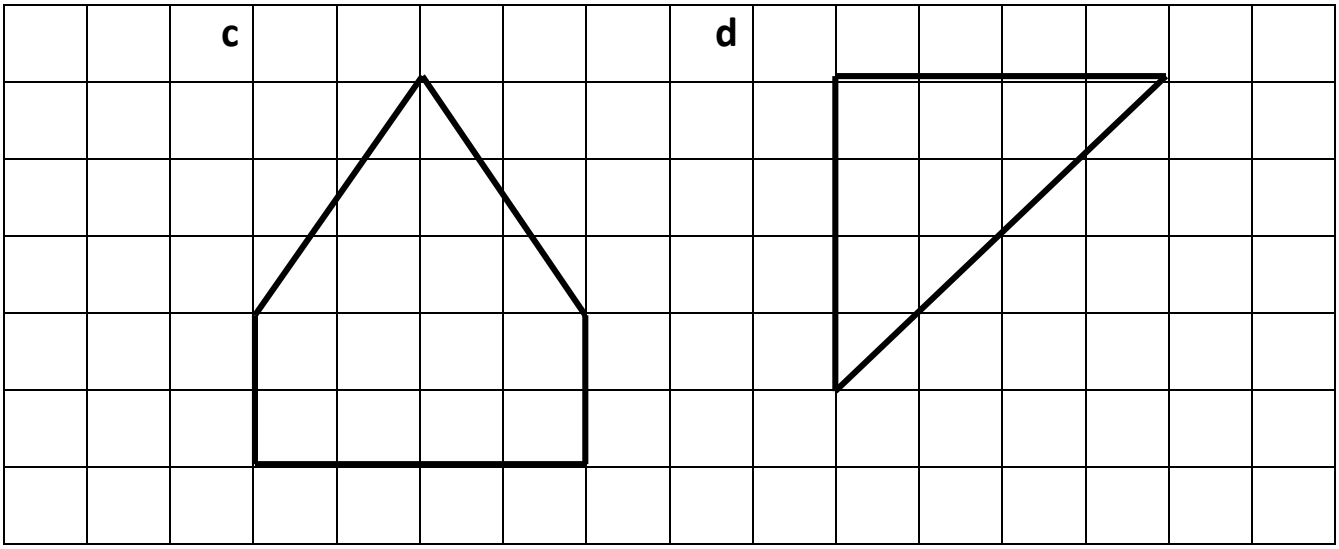


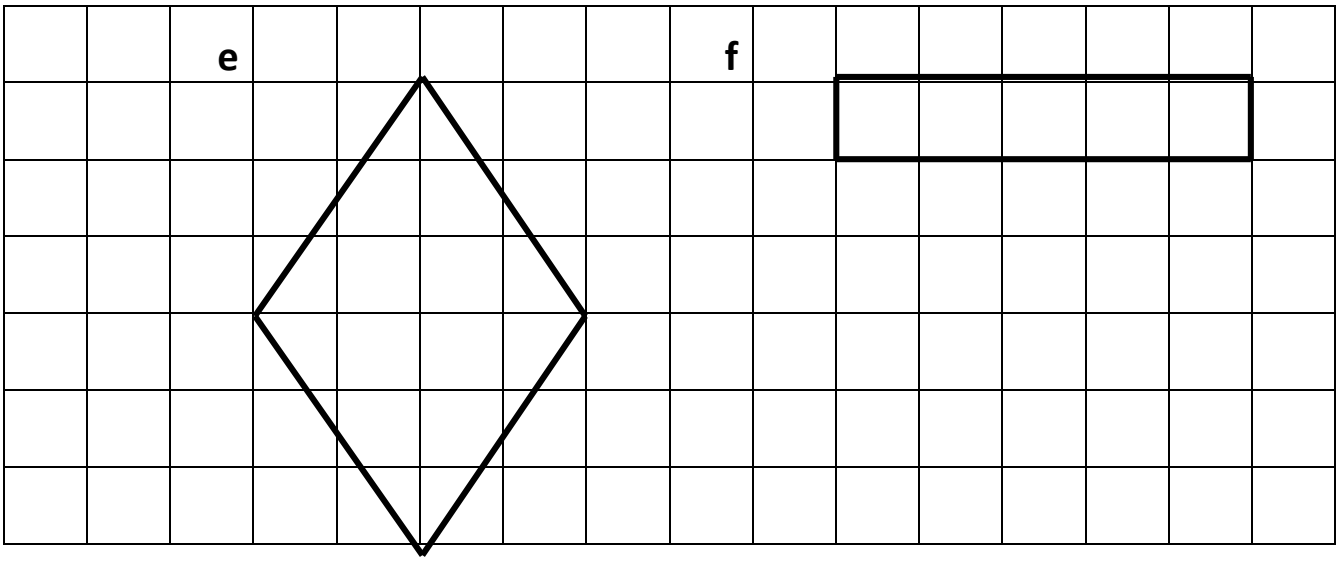
4)

Use the scale 1 cm = 4 metres to calculate the areas of the following



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5) A man takes 200 paces around a square. If each pace is one meter, what is the area of the square?

6) A drawing is scaled down in the ratio 1:10. On the drawing the sides of a rectangle are 10 cm by 15 cm. What is the area of the full size rectangle?

7) The area of a square is 2500 cm^2. If the sides of the square are reduced to one fifth their size, what is the area of the new square?

8) A movie projector blows the picture on a screen up to one hundred times its original size. What area would a triangle of base length 2 cm and height 5 cm have on the projector screen?

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Exercise 3

Volume & Capacity

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1) How much water is displaced from a container if a cube with the following side lengths is placed into it?

a) 10 cm

b) 20 cm

c) 40 cm

d) 1 m

2) What is the side length of a cube that displaces the following amount of water?

a) 125 litres

b) 8 litres

c) 27 litres

d) 1000 litres

3) How much water is displaced from a container if a cube with the following side lengths is placed into it?

a) 1 cm

b) 7 cm

c) 5 cm

d) 3 cm

4) What is the side length of a cube that displaces the following amount of water?

a) 1 mL

b) 64 mL

c) 512 mL

d) 27 mL

5) A rock having a volume of 805 cubic centimetres is placed in a full tub of bath water. How much water is displaced?

6) Seven litres of water is displaced from a full swimming pool when a large object is placed into it. What was the volume of the object?

7) Archimedes reputedly was the first person to realise the relationship between the volume of a body being placed in water and the amount of water that is displaced. He is alleged to have been so excited that he immediately leapt from his bath shouting "Eureka" when he realised his discovery. If Archimedes had a volume of 0.095 m^3, how much water would he have displaced from his bath?

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8) Convert the following to litres

a) 1400 mL

b) 3500 mL

c) 2225 mL

d) 3950 mL

e) 5010 mL

9) Convert the following to mLs

a) 2.12 litres

b) 1.356 litres

c) 0.1 litres

d) 0.87 litres

e) 10.01 litres

f) 0.01 litres

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Exercise 4

Mass

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1) A sack of identical rocks weighs 54 kg. If there are 9 rocks in the bag, what is the mass of each rock?

2) A packet of lollies weighs 400 g. If there are 20 lollies in the pack, how much does each lolly weigh?

3) The mass of an empty container is 40 grams. Into the container there are placed 15 identical wooden blocks. The container is put on a scale and it reads 265 grams. How much does each block weigh?

4) A container has a mass of 20 grams. When some identical pens are put into it the mass increases to 220 grams. If each pen weighs 40 grams, how many pens are placed in the container?

5) A container holding twelve identical pieces of plastic each weighing 11 grams is placed on a scale which reads 170 grams what is the mass of the container?

6) Convert the following to grams

a) 1 kg

b) 4 kg

c) 6.5 kg

d) 3.25 kg

e) 0.932 kg

f) 1.604 kg

7) Convert the following to kg

a) 1000 g

b) 3000 g

c) 2500 g

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d) 875 g

e) 10001 g

f) 203 g

8) What is the mass of the following quantities of water?

a) 1 litre

b) 5 litres

c) 3.5 litres

d) 0.75 litres

e) 400 mL

f) 2500 mL

9) What is the volume of the following masses of water?

a) 1 kg

b) 4 kg

c) 2.5 kg

d) 0.6 kg

e) 300 g

f) 4250 g

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Exercise 5

Time

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1) The following shows Peter's school timetable

	Monday	Tuesday	Wednesday	Thursday	Friday
9 – 9:55	Maths	Science	Option 1	SOCE	Option 2
10-10:55	English	Science	Maths	Science	Option 2
11-11:55	Computing	Option 2	Maths	Computing	English
12-12:35	Lunch	Lunch	Lunch	Lunch	Lunch
12:40-1:35	SOCE	Maths	Science	Maths	SOCE
1:40-2:35	SOCE	SOCE	English	Health	Sport
2:40-3:35	Option 1	English	English	Health	Sport

a) How long does each class last for?

b) How many hours of computing does peter have per week?

c) Peter chose woodwork as his second option. When does he have woodwork class?

d) Peter likes maths & English but doesn't like SOCE. Which day is probably his favourite day at school?

e) How many subjects have "double periods"?

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2) The following timetable shows the places and departure and arrival times of free buses to "The Big Day Out" music festival



FREE BUS TIMETABLE

BUS 1

Ventnor, Boots Bus Stop	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Niton, Church Street	10:20	Brighstone, School	15:50
Brighstone, School	10:40	Niton, Church Street	16:10
Freshwater, Big Day Out	11:00 Arrive	Ventnor, Boots Bus Stop	16:30 Arrive

BUS 2

Bembridge Library	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Sandown, High Street	10:20	Sandown, High Street	16:20
Freshwater, Big Day Out	11:10 Arrive	Bembridge Library	16.40 Arrive

BUS 3

Shanklin, Bus Station	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Freshwater, Big Day Out	11:00 Arrive	Shanklin, Bus Station	16.30 Arrive

BUS 4

Ryde Commodore	10:00 10:30 Depart	Freshwater, Big Day Out	15:30 16:00 Depart
Freshwater, Big Day Out	11:00 11:30 Arrive	Ryde Commodore	16.30 17:00 Arrive

BUS 5

Cowes Library	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Freshwater, Big Day Out	10:50 Arrive	Cowes Library	16.20 Arrive

BUS 6

Newport Library	10:00 10:30 Depart	Freshwater, Big Day Out	15:30 16:00 Depart
Freshwater, Big Day Out	10:30 11:00 Arrive	Newport Library	16.00 16:30 Arrive

BUS 7

East Cowes Town Hall	10:00 Depart	Freshwater, Big Day Out	15:30 Depart
Freshwater, Big Day Out	10:45 Arrive	East Cowes Town Hall	16.15 Arrive

a) How many buses are there?

b) Which bus runs from Cowes Library?

c) How many places does bus 1 pickup from?

d) How long does the trip from The Big day Out to East Cowes Town Hall take?

e) How long does it take to get from Sandown to Bembridge Library?

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3) The timetable for three trains from Midland to Perth is shown below

	a.m.	p.m.	
Midland	8:35	2:28	4:06
West Midland	Exp	2:30	4:08
East Guildford	Exp	2:32	4:10
Guildford	Exp	2:33	4:11
Success Hill	Exp	2:34	4:12
Bassendean	8:40	2:36	4:14
Ashfield	Exp	2:38	4:16
Bayswater	8:43	2:41	4:19
Meltham	8:45	2:43	Exp
Maylands	8:46	2:44	Exp
Mt. Lawley	8:48	2:46	Exp
East Perth	8:50	2:48	Exp
Claisebrook	8:51	2:49	4:23
McIver	8:53	2:51	4:25
Perth	8:55	3:00	4:27

Exp = Express

a) Which train gets you to Perth the quickest?

b) How long does it take to get from Bassendean to East Perth on a non express train?

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c) How much time is saved between Bayswater and McIver if the express train is taken?

d) Name a station that all three trains stop at

e) How long does an express train take to get from Midland to Bassendean?

4) The timetable below shows air arrivals and departures to and from Gibraltar

Information kindly supplied by Brian T Richards.

Day	Flight No.	Airline	Arr.	From	Dep.	Flight No.	To
Monday	ZB068	Monarch	11.00	Luton	12.00	ZB069	Luton
	ZB574	Monarch	11.15	Manchester	12.00	ZB575	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.50	Gatwick	14.25	EZY8904	Gatwick
Tuesday	ZB068	Monarch	11.55	Luton	12.40	ZB069	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.40	Gatwick	14.10	EZY8904	Gatwick
Wednesday	ZB574	Monarch	11.00	Manchester	11.45	ZB575	Manchester
	ZB068	Monarch	11.00	Luton	12.00	ZB069	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.50	Gatwick	14.25	EZY8904	Gatwick
Thursday	ZB068	Monarch	10.40	Luton	11.30	ZB069	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.50	Gatwick	14.25	EZY8904	Gatwick
Friday	ZB574	Monarch	11.00	Manchester	11.45	ZB575	Manchester
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	EZY8903	easyJet	13.45	Gatwick	14.25	EZY8904	Gatwick
	ZB062	Monarch	18.55	Luton	19.55	ZB063	Luton
Saturday	EZY8901	easyJet	11.05	Gatwick	11.40	EZY8902	Gatwick
	ZB068	Monarch	11.55	Luton	12.40	ZB069	Luton
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
Sunday	EZY8901	easyJet	11.05	Gatwick	11.40	EZY8902	Gatwick
	BA490	British Airways	11.55	Heathrow	12.45	BA491	Heathrow
	ZB062	Monarch	18.10	Luton	18.55	ZB063	Luton

a) What time does the plane to Gatwick leave on Tuesday?

b) How many times a week does easy Jet fly into Gibraltar?

c) Name an airport that Monarch does not fly into

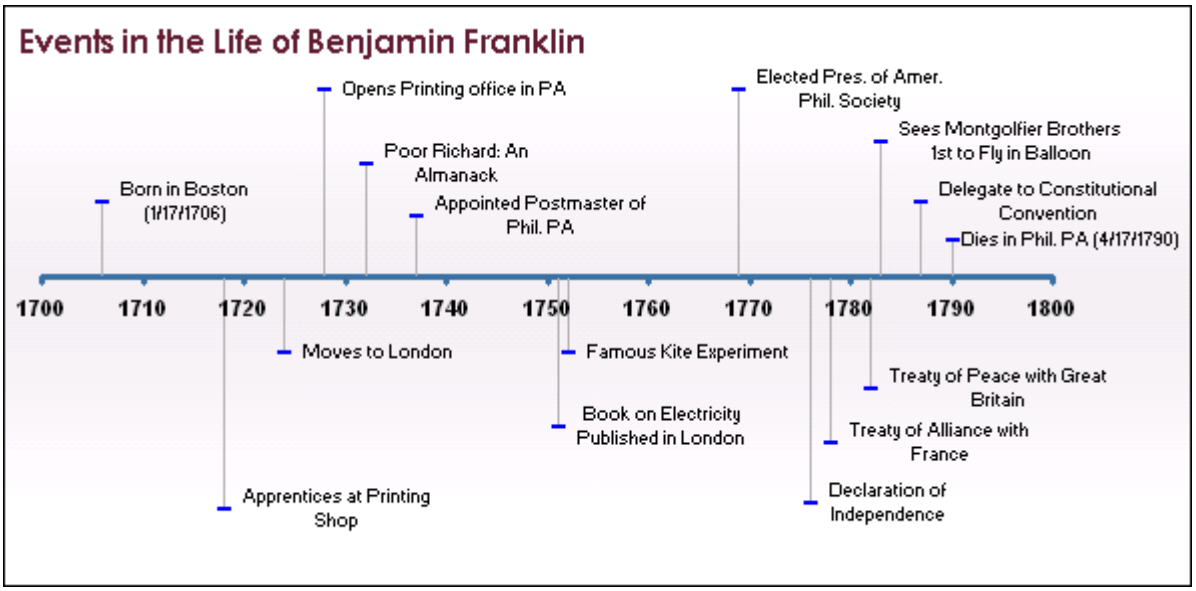
d) Which airline uses Heathrow?

e) On Friday how long does the flight that lands at 13:45 sit on the tarmac before it takes off again?

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f) How many times a week does flight ZB575 fly out of Gibraltar?

5) The timeline shows the major events in the life on Benjamin Franklin



a) How long did Benjamin Franklin live for?

b) In approximately what year did he move to London?

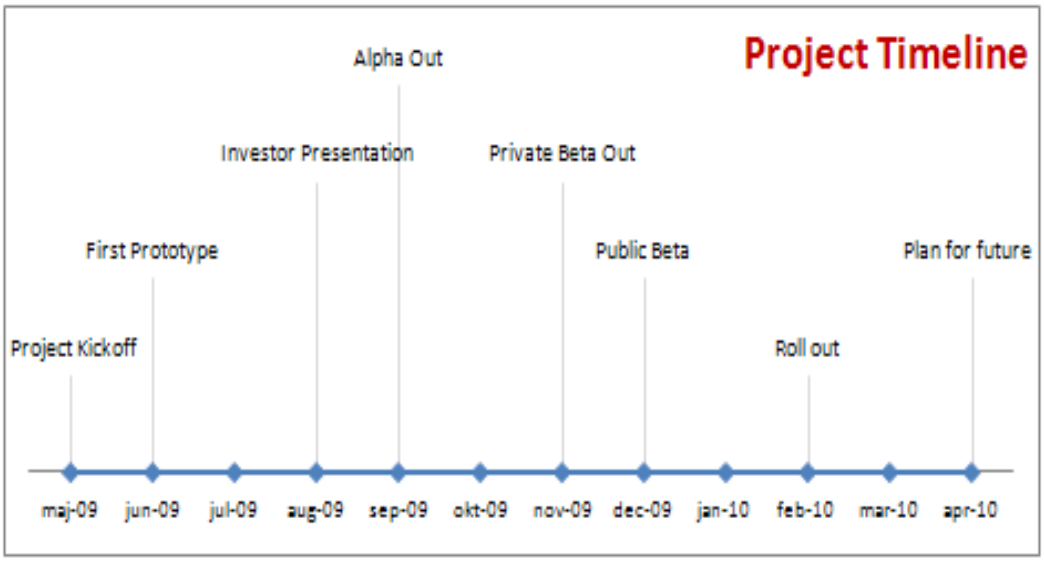
c) How old was he when the Declaration of Independence was signed?

d) Approximately how many years after he published “Poor Richard” did he perform his famous kite experiment?

e) In approximately what year was he elected President of the American Philosophical Society?

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6) A project is planned from start to finish by the timeline below



a) When did the project start?

b) How long from start up to roll out?

c) When was the investor presentation?

d) How long did the project allow for development of the first prototype?

e) How long after the roll out were future plans made?

7) Construct a timeline for a typical day from 6am to 8pm for a year 6 student on a school day

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8) Put the following events on a suitably scaled timeline

Technology Timeline 1400-1890

1450-Printing Press Invented
1543-Copernicus Claims Earth Circles the Sun
1543-First Illustrated Study of Human Body
1609-Kepler Publishes His Laws of Planetary Motion
1638-Galileo Explains Principles of Falling Bodies
1669-Newton Explains Calculus
1698 Steam Pump Introduced
1733 Flying Shuttle Invented
1764-Spinng Jenny Introduced
1776-First Submarine Launched
1776-Watt Builds Steam Engine
1784 Power Loom Invented
1793-Cotton Gin Invented
1814-George Stephenson Develop Steam Engine
1819-First Steam Powered Ship Crosses Atlantic
1829-Baltimore and Ohio Railroad Begun
1837-Steel Plow Invented
1839-First Photo
1860-Work Begun on London Underground Railroad
1876-Telephone Invented
1879-Electric Light Invented
1888-George Eastman Patents Camera
1890-Eiffel Tower

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Year 6 Mathematics

Space

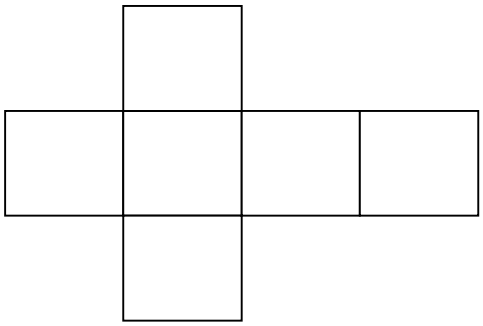
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Useful formulae and hints

When drawing views of stacks of cubes from different sides, depth is not shown. Only draw what would be seen from directly in front of the stack

The net of a shape is the 2D representation that can be folded and/or cut to make a 3D shape

The net of a cube is



Angles are classified according to their size

For whole numbers:

0-89 degrees: acute

90 degrees: right

91-179 degrees: obtuse

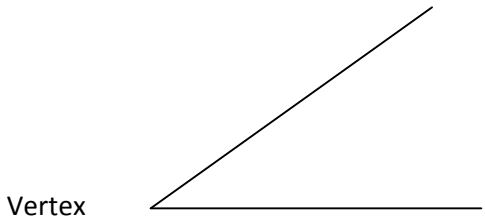
180 degrees: straight

181-359 degrees: reflex

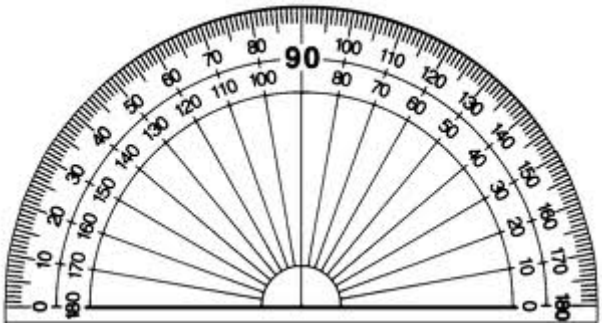
360 degrees: revolution

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Angles are measured from the point where the arms of the line segments making up the angle meet (the vertex). Angles are measured using a protractor



A protractor has two scales; the top scale measures angles that open toward the left, and the bottom measures those angles opening toward the right. Note that the sum of the two scales is always 180 degrees.



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Exercise 1

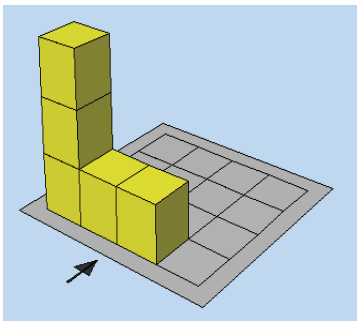
Representation of 3D Shapes

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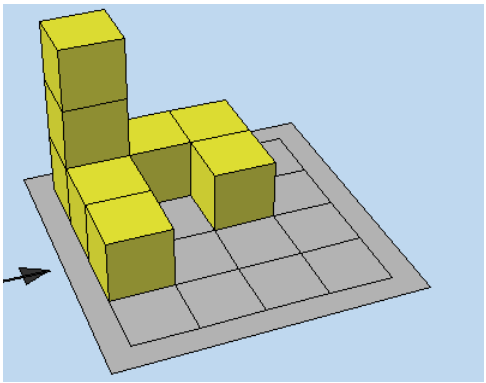
1) Draw the net of the cube shown below



2) Draw the front and two side views of the following shape (the arrow indicates the front)

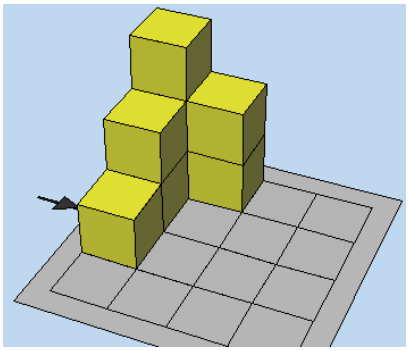


3) Draw the front and two side views of the following shape (the arrow indicates the front)



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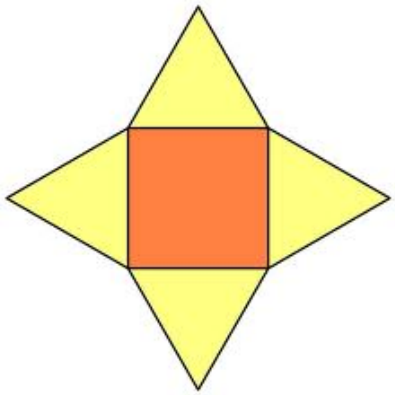
4) Draw the front and two side views of the following shape (the arrow indicates the front



5) Show the isometric representation of a cube

6) Draw the net of a triangular prism

7) The following is the net of which shape?



8) Draw the six possible nets for a square pyramid

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Exercise 2

Angles

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1) Draw the following angles

a) 45°

b) 90°

c) 60°

d) 30°

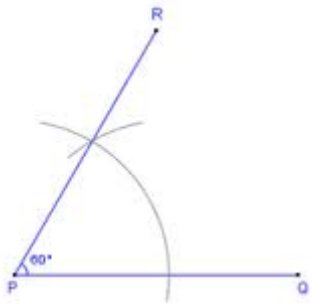
e) 100°

f) 125°

g) 180°

2) Measure the following angles

a)

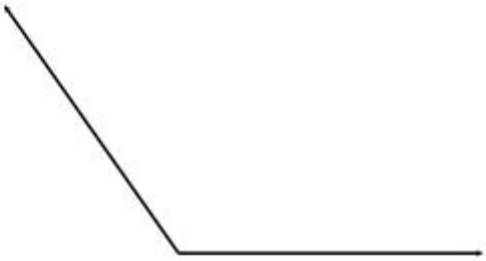


b)



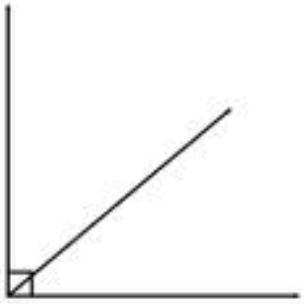
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c)



d)

e)



3) Classify the following angles

a) 30°

b) 123°

c) 180°

d) 10°

e) 90°

f) 359°

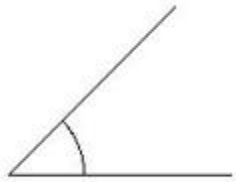
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g) 147°

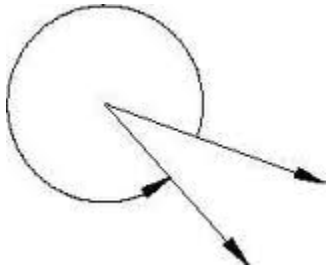
h) 360°

4) Identify the angle type formed by the intersection of the lines

a)



b)



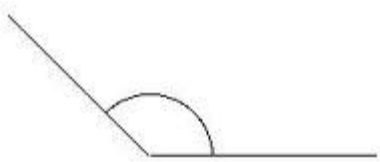
c)

d)

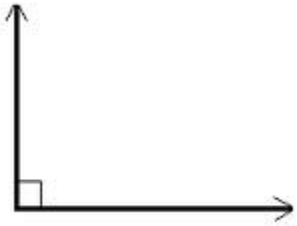


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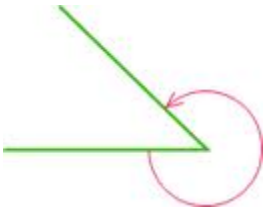
e)



f)



g)



5) What is the largest (whole number) value that can be added to each of the following angles so that they remain acute?

a) 30°

b) 42°

c) 10°

d) 57°

e) 88°

f) 0°

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6) What is the smallest (whole number) value that can be added to the following angles to make them obtuse?

a) 26°

b) 11°

c) 0°

d) 89°

e) 92°

7) What is the smallest (whole number) value that can be added to the following angles to make them reflex?

a) 23°

b) 93°

c) 0°

d) 126°

e) 178°

f) 222°

8) What is the largest (whole number) value that can be added to the following angles to make the largest obtuse angle possible?

a) 23°

b) 93°

c) 0°

d) 126°

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e) 178°

f) 22°

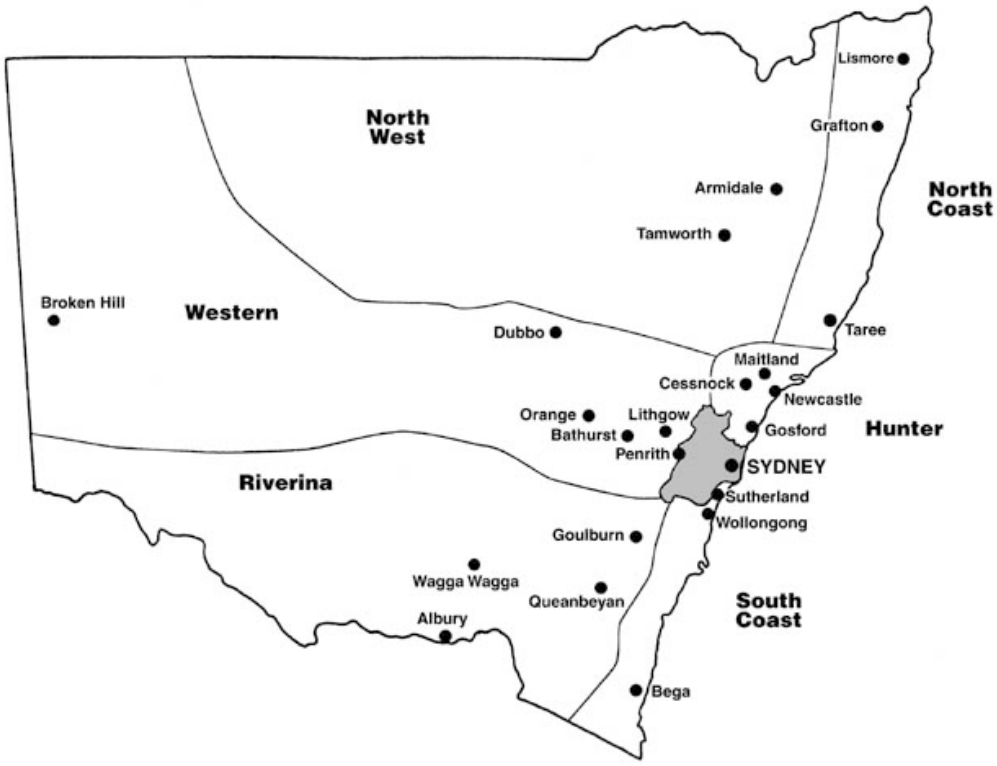
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Exercise 3

Using Maps

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1) Use the map of NSW below to answer the following questions



a) Which direction is Lismore from Grafton?

b) Which is the westernmost town on the map?

c) Name two towns in the Riverina district

d) Which town is closest to Orange?

e) Victoria is south of NSW; which town lies on the border of the two states?

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2) Answer the questions from the following map of Trinidad & Tobago



a) Which of the two islands is bigger?

b) Approximately how wide is Trinidad?

c) What is the most north easterly town of Trinidad?

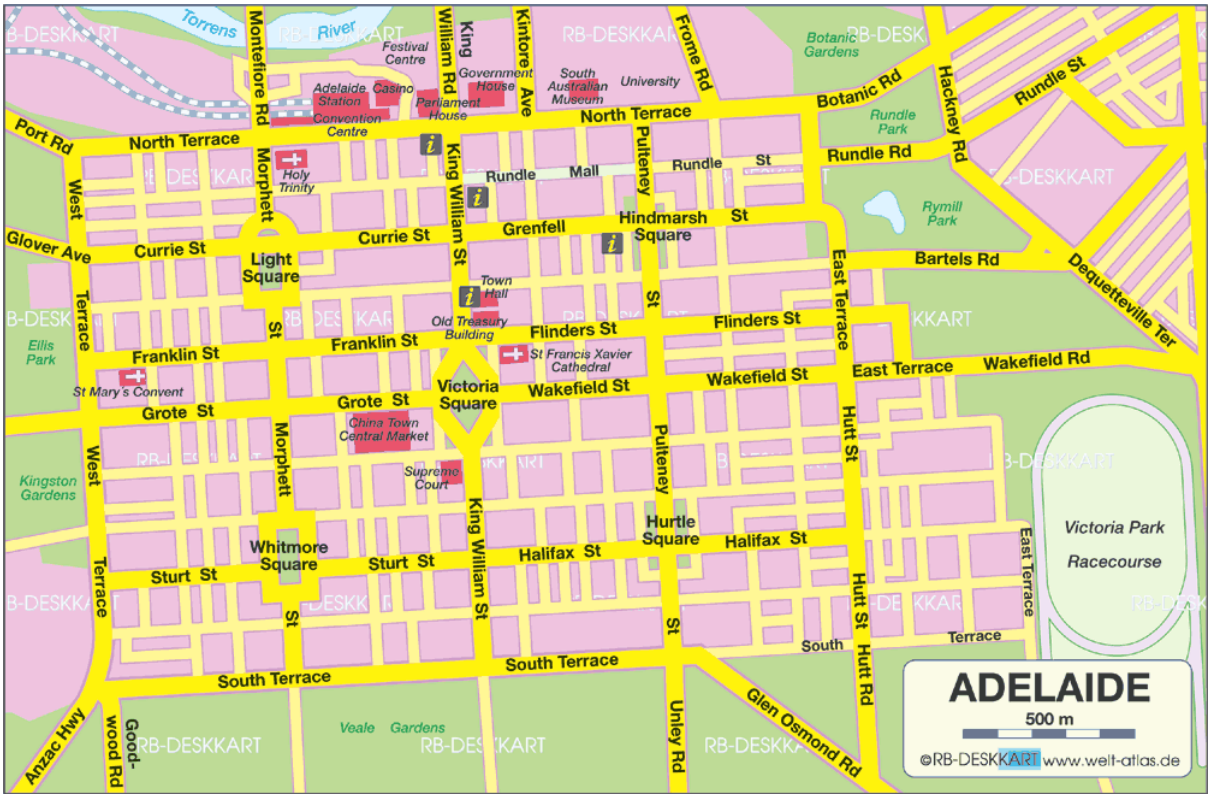
d) What sea separates the two islands?

e) What is the approximate distance from Port of Spain to Scarborough?

f) What town lies due south of Sangre Grande?

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3) Use the street map of Adelaide city to answer the questions



a) What direction is Light Square from Victoria Square?

b) How long is South Terrace?

c) Franklin Street becomes what street?

d) What is the approximate walking distance from the corner of Halifax and King William streets to Whitmore Square?

e) Which direction does Glen Osmond drive run?

f) At which end of Adelaide is Victoria Park racecourse?

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4) Using the scale 1cm = 10 km, show the following information on a map

- Boomtown is 50 km due east of Alphaville
- Cooltown is 6 km south east of Boomtown
- Dogville is 20 km due east of Cooltown
- Eastborough is 30 km south west of Dogville
- Finishton is both due west of Eastborough and due south of Alphaville

5) A factory site has a main office and six workshops. Each workshop is the same distance from the office. Each workshop is exactly in between its neighbors. Show this information on a map.

6) Fill in the map below with the names of the states and capital cities of each one. Use a suitable scale to indicate distances, and supply some written information on relative locations (e.g. Perth is x km west of Sydney)



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